Spatiotemporal Analysis of Coastal Seagrass Dynamics in the Upper Florida Keys

Final Project

04/21/2026

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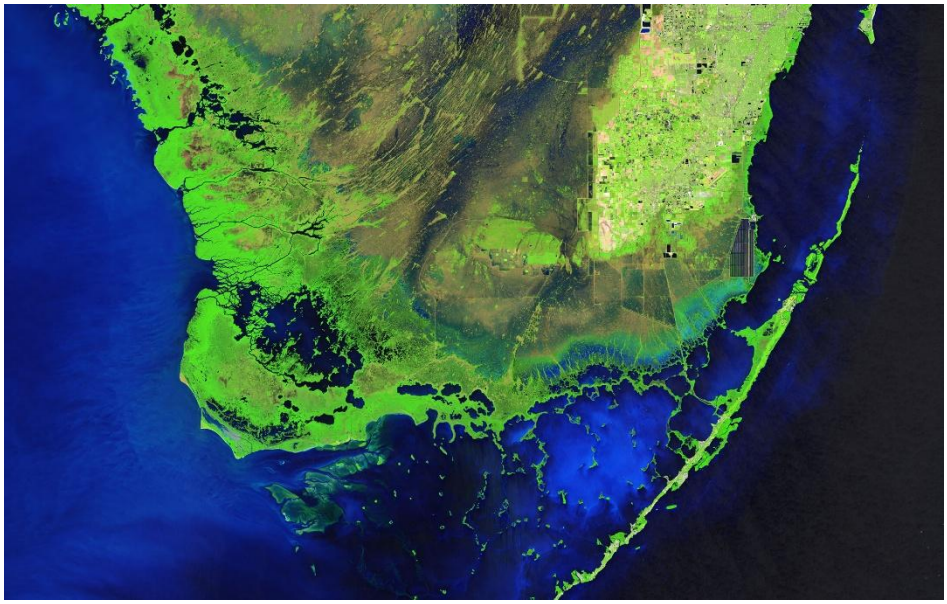

Figure 1. *Landsat Vegetation Map (FIU)*

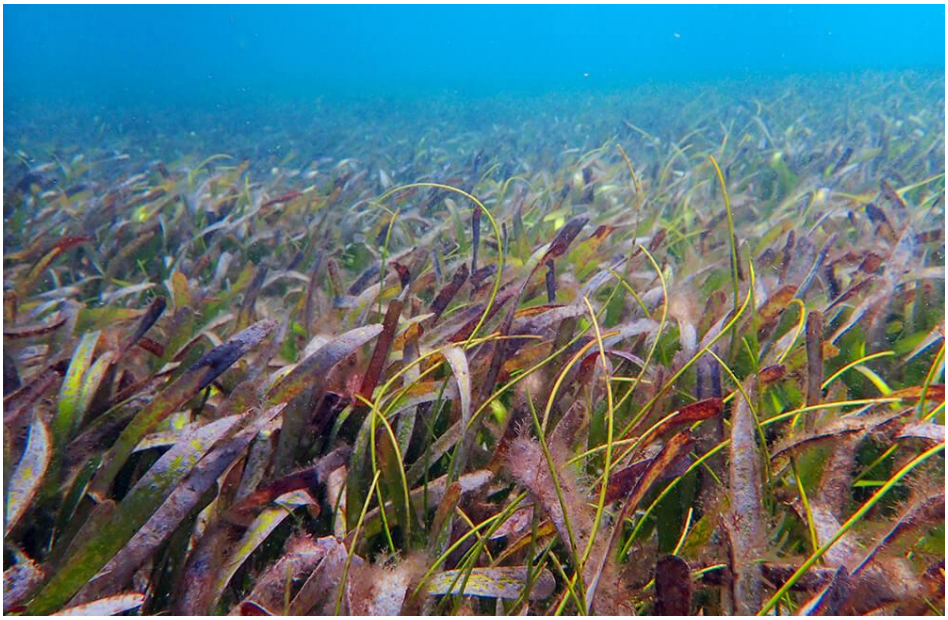

Figure 2. *Seagrass meadow (NOAA)*

Abstract

This study investigates the relationship between seagrass distribution and key environmental drivers in the Upper Florida Keys over a ten-year period, with particular emphasis on water quality conditions such as salinity, temperature and turbidity, a key determinant of water clarity and light availability. The analysis integrates Sentinel-2 multispectral satellite imagery with in-situ water quality data obtained from the Water Quality Portal. Spatial analysis is conducted within the Upper Florida Keys region, with temporal coverage spanning from 2015 to 2024 to capture both short-term variability and long-term environmental trends. Data processing and analysis are implemented using Python-based geospatial and numerical libraries. Results are presented through a combination of visualizations and time-series analyses to illustrate patterns in water quality variability and seagrass distribution across the study area. Data is limited to observable environmental drivers of seagrass distribution and does not include direct field data.

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Acronyms and Abbreviations Used

S2 - Sentinel-2
FL – Florida
AOI – Area of Interest
SCL – Scene Classification Layer
NOAA - National Oceanic and Atmospheric Administration
EPA - Environmental Protection Agency
USGS - United States Geological Survey
IEA - Integrated Ecosystem Assessment
GEMM - Groundwater Exchange Monitoring and Modeling plan for Central Florida Bay
FKNMS - Florida Keys National Marine Sanctuary
FIU - Florida International University
SECAR - Statewide Ecosystem Assessment of Coastal and Aquatic Resources
WQP - Water Quality Portal [6]
ARSET – Applied Remote Sensing Training Program (NASA)

Spectral Indices:

NVI - Normalized Vegetation Index
NDVI - Normalized Difference Vegetation Index
NDVAI - Normalized Difference Aquatic Vegetation Index
EVI - Enhanced Vegetation Index
NDWI - Normalized Difference Water Index
NDTI - Normalized Difference Turbidity Index
DII - Depth Invariant Index

GitHub

https://github.com/ssullivan9661/seagrass-analysis/tree/main

1. Introduction

Seagrasses are submerged flowering plants that grow in shallow coastal waters where sufficient light reaches the seafloor. Unlike seaweed (marine algae), seagrass are flowering plants with true roots, leaves, and vascular structures that allow them to anchor in sediment and transport nutrients and water through the plant. These ecosystems form dense underwater meadows that rank among the most productive coastal ecosystems globally, providing critical nursery habitats for a wide range of marine species, including ecologically important organisms such as the spiny lobster, bonefish, and juvenile reef fishes. In addition to supporting biodiversity, seagrass meadows play a vital role in stabilizing sediments, improving water clarity through particle trapping, cycling nutrients, and sequestering carbon, making them important contributors to both coastal resilience and climate regulation [15].

The Upper Florida Keys represent one of many ecologically significant and environmentally sensitive coastal regions in the southeastern United States. This region forms a transitional zone between the Atlantic Ocean and the adjacent Florida Bay — Everglades estuarine system and contains extensive seagrass meadows that form the ecological foundation of shallow-water habitats. However, these systems have experienced substantial environmental stress in recent decades. A major seagrass die-off event in 2015 resulted in the loss of approximately 40,000 acres of seagrass in Florida Bay, driven by a combination of elevated turbidity, hyper-salinity, nutrient enrichment, and prolonged drought conditions [2]. This event marked a major ecological shift and provides a critical baseline for evaluating long-term ecosystem recovery and resilience.

Seagrass distribution in this region is strongly influenced by environmental conditions, particularly water clarity. Light availability is the primary limiting factor for seagrass growth, and increased turbidity caused by suspended sediments, algal blooms, and dissolved organic matter reduces light penetration to the seafloor, directly constraining photosynthesis and harming ecosystem productivity [3]. Sentinel-2 multispectral imagery enables high-resolution observation of coastal vegetation dynamics, while in-situ water quality data allow for assessment of environmental drivers. This study integrates S2 imagery (2015–2024) with water quality observations from the Water Quality Portal (WQP) to examine how turbidity and related water quality parameters influence seagrass distribution across the Upper Florida Keys.

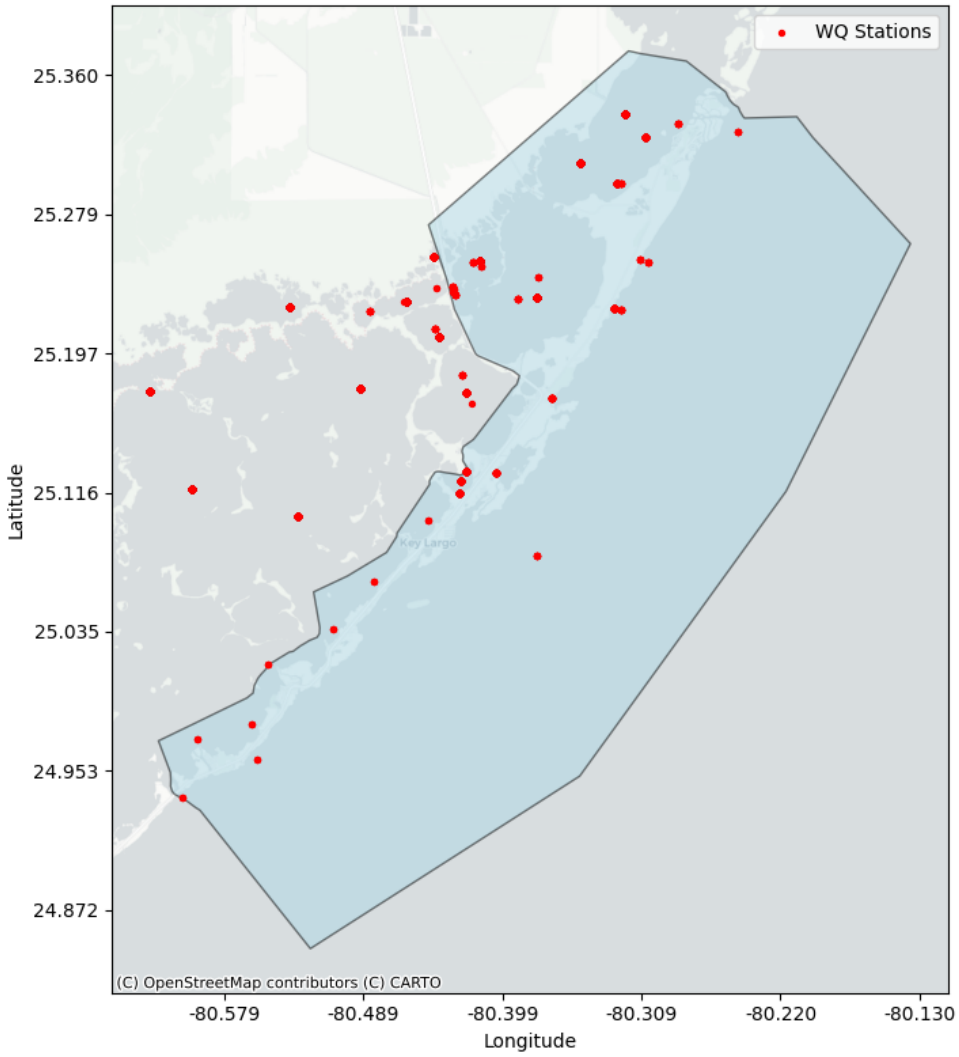
2. Methods

This study utilizes two primary data sources: S2 multispectral satellite imagery obtained from the Copernicus Data Space Ecosystem and water quality observations collected by the USGS and EPA via the WQP. S2 offers publicly accessible satellite imagery at 10-meter spatial resolution with a five-day revisit cycle, allowing for consistent monitoring of submerged vegetation dynamics and coastal water conditions.

S2 imagery was programmatically searched and retrieved using GeoJSON bounding box coordinates of the study area, the Upper Keys, Fl. Search parameters for Copernicus Data Space Ecosystem, included resolution, percent cloud cover, classification levels L1C (2015-2016) and

L2A (2017-2024), due to changes in global/national product standards of S2 overtime. Additionally, the pipeline queries and retrieves data from the WQP, organized and filtered for validated historical records of turbidity, temperature, salinity, and Secchi depth.

Figure 3. Bounded study area. Red dots represent monitoring stations that reported relevant data to the WQP (2015-2024). Created in Python with Matplotlib, Pandas, GeoPandas, and Contexily.



2.1 Preprocessing

Preprocessing of S2 imagery was conducted to ensure spatial and temporal consistency across the full dataset. This included scene selection, atmospheric consistency through L2-A and L1-C products, and cloud masking using the S2 Scene Classification Layer (SCL) to remove clouds, cloud shadows, and invalid pixels for L2A products. Each scene was then clipped to the defined Area of Interest (AOI) using geospatial boundaries derived from a shapefile.

2.1.1 Preprocessing Implementation

All processing was conducted in Python using libraries such as GeoPandas and Shapely for spatial handling, and Rasterio and RioXarray for raster processing. Each scene was unzipped, and converted from JP2 → TIFF files as multi-dimensional Data Arrays with Xarray, then saved as netCDF. By saving as netCDF, the project attempts to utilize lazy loading, to allow the system to easily read metadata including coordinates without pulling all 145 scenes into memory.

Demonstrating computational efficiency using a "memory-first" strategy that leverages Xarray Data Arrays to handle computationally heavy S2 raster band data and once preprocessed and saved, raw version is purged from memory. All scenes and WQP data were reprojected to the study area Coordinate Reference System (CRS) with GeoPandas. Cloud masking was applied via the SCL for all L2A level scenes with RioXarray, specifically for the filter classes 3, 8, 9, and 10 to ensure benthic visibility. While L1C utilized `masked=True` to handle the NoData values in the process.

2.2 Spectral Indices

A set of spectral indices were calculated from S2 imagery to characterize aquatic vegetation presence and coastal water quality conditions relevant to seagrass distribution. The Normalized Difference Aquatic Vegetation Index (NDAVI) was used as the primary proxy for submerged vegetation in shallow coastal waters, providing improved sensitivity to seagrass detection compared to traditional vegetation indices [12, 13]. The Normalized Difference Water Index (NDWI) was applied to distinguish open water from vegetated or submerged surfaces, and the Normalized Difference Turbidity Index (NDTI) was used to estimate relative water clarity and suspended sediment concentrations, which directly influence light availability for photosynthesis [13].

2.2.1 Spectral Index Calculation

Calculations of spectral indices post preprocessing were executed using Xarray and NumPy, utilizing the following spectral bands.

Band	**Associated indices**	**Importance for project**
Blue	NDWI, NDTI, SSI-II	Short wavelength, low absorption; Penetrates farthest in water.
Green	NDWI, NDTI, SSI-II	Highly reflectance with longer wavelength, allows to see vegetation or water clarity.
Red	EVI, NDVI, NDTI	Long wavelength, high absorption; Less visibility of benthos but strong chlorophyll-a absorption.

NIR (Near Infrared)	EVI, NDVI, NDWI, NDAVI	High reflectance of healthy vegetation.

[9],[10],[11],[12].

Calculations for NDVI, NDAVI, EVI, NDTI, and SSSII were sourced from NASA's ARSET[13] (1-5). Calculations for NDWI and DII are from Gao[10] (6)and Lyzenga[14] (7). The spectral index equations mentioned are:

$$NDVI = \frac{(NIR - Red)}{(NIR + Red)}$$

(1)

$$NDAVI = \frac{(NIR - Blue)}{(NIR + Blue)}$$

(2)

$$EVI = \frac{2.5 * (NIR - Red)}{(NIR + (6) * Red - (7.5) * Blue + 1)}$$

(3)

$$NDTI = \frac{(Red - Green)}{(Red + Green)}$$

(4)

$$SSI - II = \frac{(Green - Red)}{(Green + Red)}$$

(5)

$$NDWI = \frac{(Green - NIR)}{(Green + NIR)}$$

(6)

$$DII = \ln\left(\frac{Green}{Blue}\right) = \ln(Green) - \ln(Blue)$$

(7)

These indices were used in tandem with in-situ water quality measurements to evaluate spatial and temporal relationships between environmental conditions and seagrass distribution.

2.3 Workflow

Temporal analysis was conducted to evaluate changes in seagrass-related spectral signals and associated water quality conditions over 2015–2024. Satellite-derived indices (NDAVI, NDWI, and NDTI) and in-situ water quality measurements were organized into structured time-series datasets to enable consistent comparison across annual and seasonal intervals. The Mann–Kendall trend test was applied to key environmental variables, including turbidity and selected spectral indices to identify the direction of significant temporal trends without assuming a

specific data distribution. Combined with time-series visualization techniques, this allowed for the assessment of long-term changes in water quality conditions and their relationship to seagrass responses across the study period.

During data integration, a temporal join of the indices summary and WQP samples is built using a 3-day tolerance window to better match in-situ recordings with days of S2 overpass. The final spatial analysis utilizes Seaborn, Cartopy, and Rasterio to communicate results; Index comparisons and visual side-by-side comparisons of NDAVI and turbidity, organized annually and seasonally, providing a direct way to validate results of the analysis and the model itself. The final output of the workflow is a high-resolution time-series animation of the Upper Keys, visualizing seagrass dynamic changes and spatial distribution of the confirmed direct seagrass indicator index, NDAVI, over the 2015-2024 study period.

3. Results & Discussion

3.1. Spatial distribution of seagrass-related spectral indices

Analysis of S2 derived indices revealed clear spatial patterns in submerged vegetation conditions across the Upper Florida Keys. The NDAVI showed higher values in shallow, coastal regions, indicating greater likelihood of seagrass presence. These areas corresponded spatially with zones of reduced turbidity indicated by the NDTI, consistent with the expectation that clearer water supports greater submerged vegetation density. The NDWI distinguished open water from submerged vegetated regions, providing clear spatial separation between benthic habitat zones and deeper channel features. Spatial mean values were computed across 145 cloud-free S2 scenes spanning 2015–2024, covering two adjacent tiles (T17RNH and T17RNJ) that together encompass the full study area extent.

3.2. In-situ Water Quality Analysis

In-situ water quality observations from 56 monitoring stations across the study period (WQP) reveal certain relationships among key environmental parameters. Pairwise analysis (Figure 4) confirms an expected inverse relationship between turbidity and Sechhi depth [11] and illustrates seasonal variability of parameters. Temperature shows a weak association with both turbidity and Secchi depth, with warmer conditions corresponding to greater variability and reduced clarity. Salinity, however, appears mostly independent of other parameters, suggesting it is driven by separate factors.

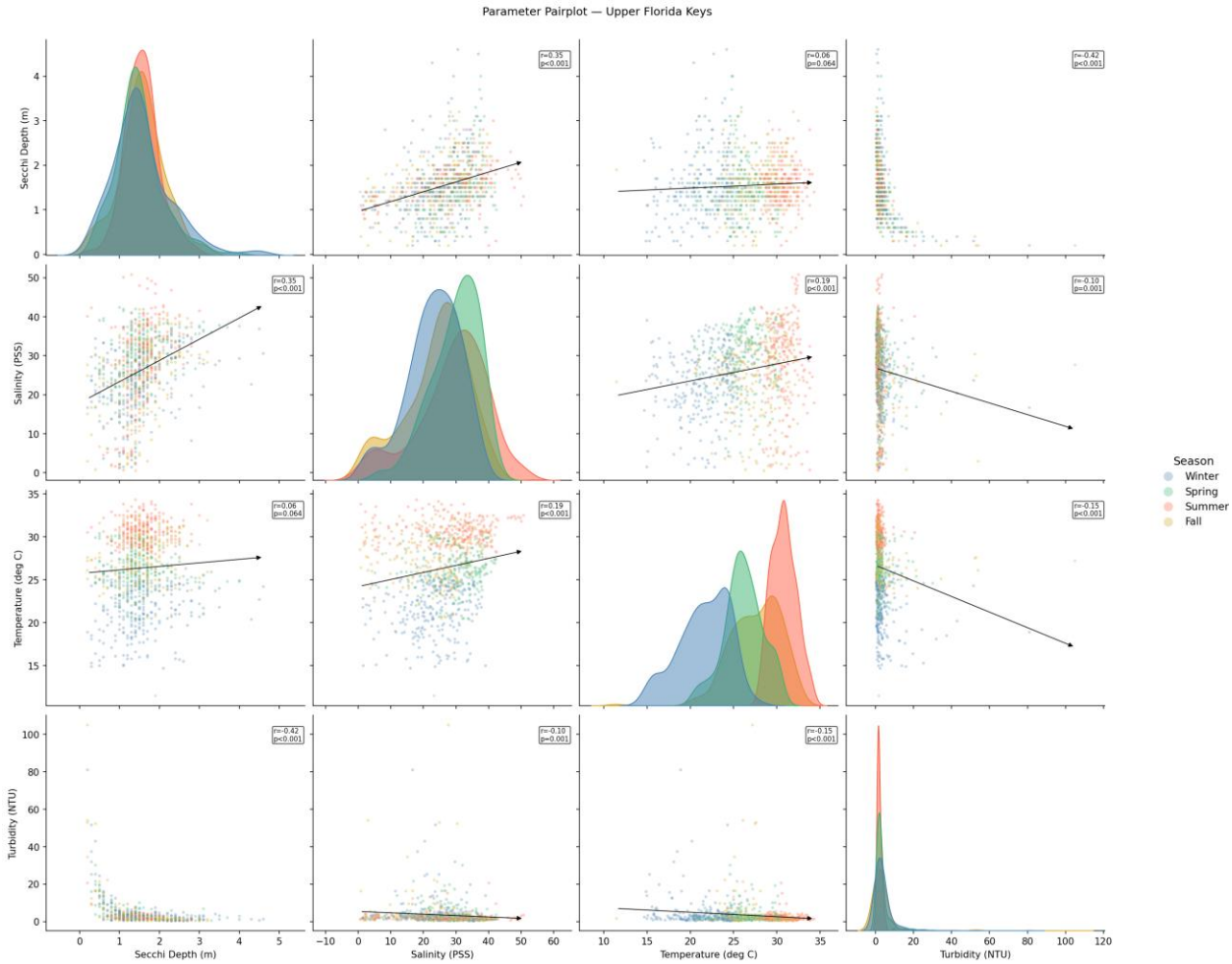


Figure 4. Pairwise relationships among turbidity, Secchi depth, temperature, and salinity, color-coded by season. Demonstrates expected inverse relationship between turbidity and Secchi depth, weak associations between temperature and clarity parameters, and independence of salinity from other variables. Created in Python with Matplotlib, Pandas, GeoPandas, and Seaborn.

3.3. Relationship between water quality and spectral response

Integration of in-situ water quality data from 56 monitoring stations with satellite-derived indices yielded 387 temporally matched observation pairs, identified using a ± 3-day tolerance window between sampling dates and satellite overpass dates. Pearson correlation analysis (Figure 5) on these matched pairs revealed statistically significant relationships between certain in-situ water quality parameters and spectral indices. Turbidity showed a significant negative correlation with NDAVI ($r = -0.288$, $p < 0.001$), confirming that higher suspended sediment amounts reduce the detectability of submerged vegetation from satellite imagery [4]. Turbidity also showed a positive correlation with NDWI ($r = 0.325$, $p < 0.001$), indicating that increased sediment amounts elevate surface reflectance in the green wavelengths that NDWI is sensitive to. Secchi Depth showed a significant positive correlation with NDAVI ($r = 0.176$, $p = 0.006$), directly supporting that improved water clarity enables greater seagrass spectral signal. Notably,

water temperature demonstrated the strongest correlations in the dataset, with SSI-II ($r = -0.669$, $p < 0.001$) and NDTI ($r = 0.669$, $p < 0.001$), indicating strong seasonal coupling between water temperature and the satellite's capacity to distinguish seagrass from bare sand substrate.

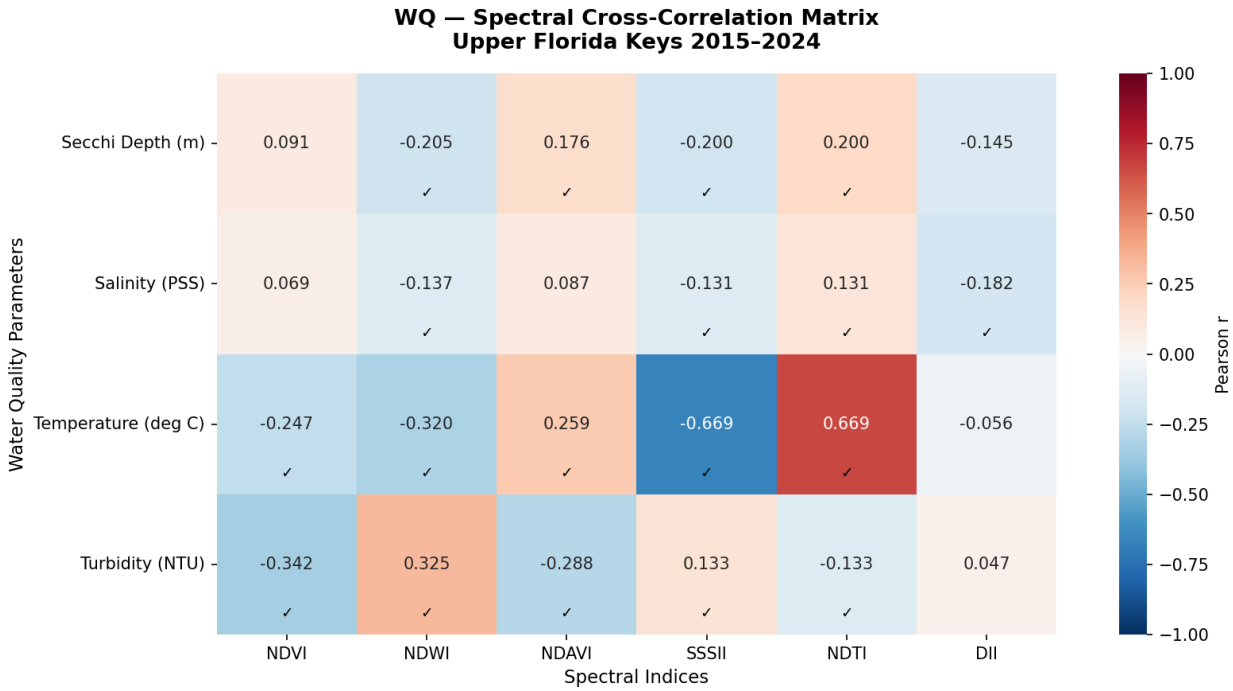


Figure 5. Pearson correlation matrix between water quality parameters and spectral indices ($n = 387$). Color intensity indicates correlation strength; significant relationships ($p < 0.05$) include turbidity-NDAVI (negative), Secchi depth-NDAVI (positive), and temperature with SSI-II and NDTI. Note exact inverse relationship between SSSII and NDTI, indicating complementary sensitivity to water clarity. Made in Python with Matplotlib, Pandas and GeoPandas.

3.4. Temporal trends in seagrass and water quality conditions

Long-term trend analysis using the Mann–Kendall test [5] revealed statistically significant improving trends in both water clarity and seagrass spectral signal over the study period (Figure 6). Secchi Depth, the primary in-situ measure of water clarity, exhibited a significantly increasing trend ($\tau = 0.163$, $p = 0.009$), indicating progressively clearer water conditions across the monitoring network. Correspondingly, NDAVI showed a significant increasing trend in satellite observations ($\tau = 0.162$, $p = 0.031$), and NDVI similarly increased significantly ($\tau = 0.153$, $p = 0.041$). The temporal coupling between Secchi depth and NDAVI (Figure 7) demonstrates concurrent improvement in both in-situ water clarity measurements and satellite-detectable seagrass, particularly following Hurricane Irma (2017) disturbance. However, no significant trends were detected in turbidity, salinity, or temperature over the same period, suggesting that the improving clarity signal is driven by factors beyond simple turbidity reduction alone.

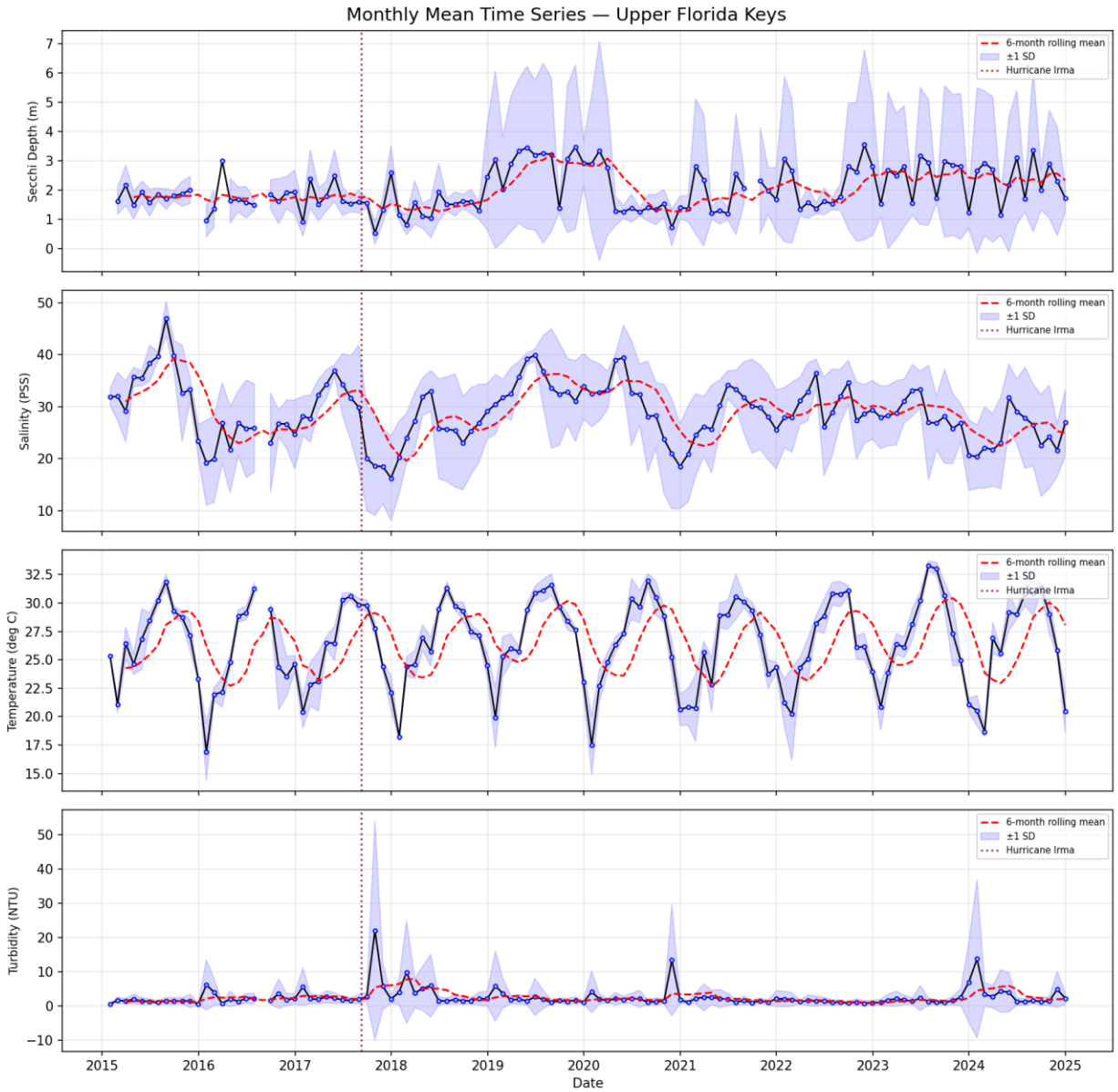


Figure 6. Mann-Kendall trend analysis for relevant parameters (2015-2024). Monthly observations (blue lines) with 6-month rolling means (red dashed lines) and ±1 standard deviation envelopes (shaded regions) show: (a) Secchi depth exhibiting slight increasing trend in water clarity, (b) Salinity displaying high variability and slight decline, (c) Temperature showing stable seasonal cycling, and (d) Turbidity characterized by sporadic extreme events. Vertical red line indicates Hurricane Irma (2017). Created in Python with Matplotlib, Pandas and GeoPandas.

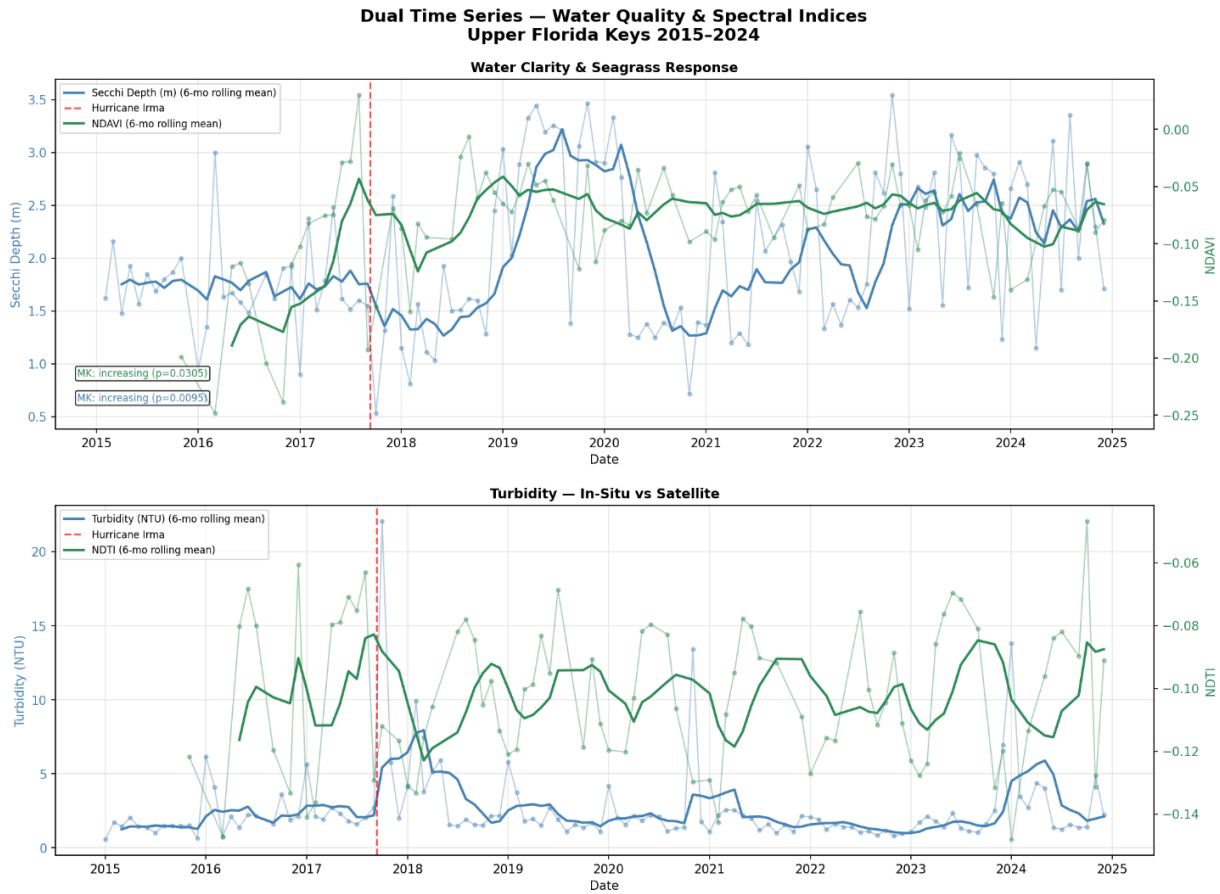


Figure 7. Dual time-series comparison of in-situ measurements and satellite-derived indices. a) Secchi depth compared against NDAVI showing coupled increasing trends in water clarity and seagrass signal. (b) Turbidity compared against NDTI shows general inconsistency. Created in Python with Matplotlib, Pandas and GeoPandas.

3.5. Anomaly Analysis

Quantifying monthly deviations from the 2015–2024 long-term mean (Figure 8) revealed a transition from predominantly negative anomalies in both Secchi Depth and NDAVI during 2015–2017 to predominantly positive anomalies from 2018 onwards. The negative anomalies observed in the early study period are consistent with the degraded water quality and vegetation loss associated with the 2015 seagrass die-off event [2], suggesting that conditions in the immediate years following the die-off remained below the average in both clarity and submerged vegetation cover. Hurricane Irma's impact in September 2017 coincides with a notable inflection point in both anomaly time series, after which Secchi Depth and NDAVI anomalies shift predominantly positive and remain so through 2024. This pattern suggests the storm event may represent an ecological turning point, potentially through resuspension of nutrient-rich sediments followed by improved water column conditions during the recovery period [3]. While causality cannot be established from the present analysis, this temporal sequence is consistent with post-disturbance recovery dynamics documented in other Florida Keys seagrass systems [7].

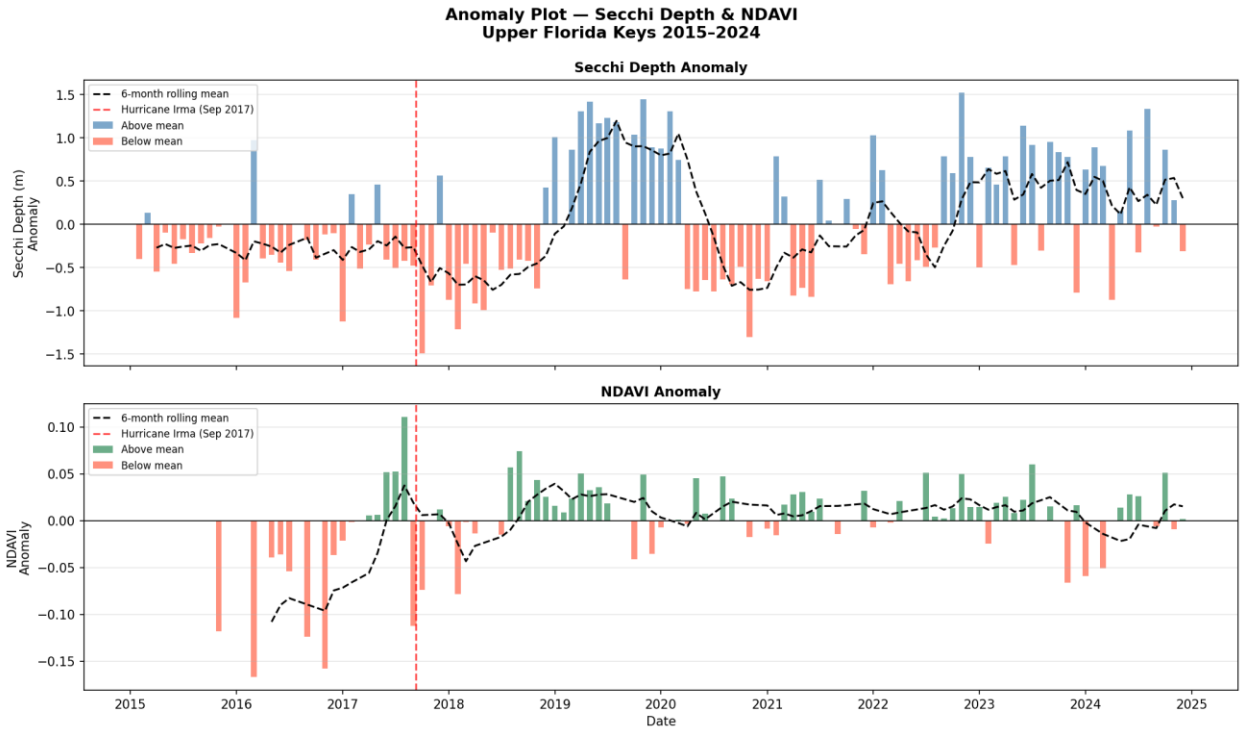


Figure 8. Standardized anomaly plot of in-situ Secchi depth measurements and satellite-derived NDAVI index (2015–2024). Concurrent positive trends suggest recovery of optical conditions and potential seagrass expansion following Hurricane Irma (2017). Created in Python with Matplotlib, Pandas and GeoPandas.

3.6. Spatial Map Analysis

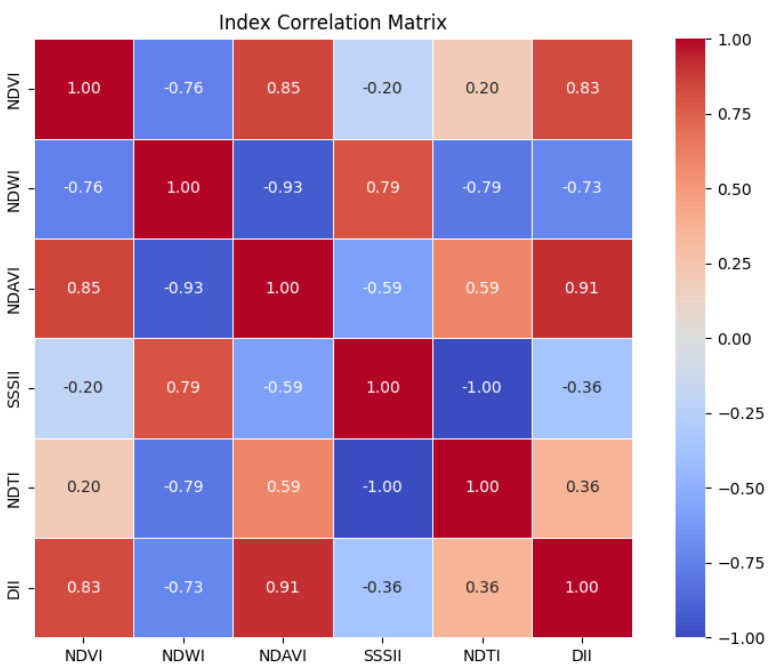


Figure 9.

Index Correlation Matrix:

A 0.91 positive correlation between NDAVI and DII confirms the accuracy of our model for NDAVI capability of tracking seagrass rather than becoming confused by refracted or reflected light. Turbidity link: A negative 0.59 indicates a high inverse relationship between NDAVI & SSSII, validating that with increased suspended sediment (turbidity) there is decrease of detectable seagrass

The spatial maps produced (Figures 11, 12) highlight the changes in seagrass distributions over time with the NDAVI index overlay. The results show more visible seagrass growth over time with a sudden drop post Hurricane Irma at the end of 2017 followed by recovery in 2018. This is consistent with the findings of the computational analysis. The spatial analysis also acted as a model validation for two reasons. For one, shown through correlation coefficient comparison between DII & NDAVI proving NDAVI is accurately detecting seagrass and not confused by reflections or floating debris. A similar study assessing spectral indices performance for detecting aquatic vegetation showed that NDAVI "enhanced capabilities" of detecting aquatic vegetation and distinguishing from land vegetation [4]. The spatial analysis further validated the inverse relationship between turbidity and seagrass distributions. Through both the negative or inverse relationship between spectral submerged seagrass index II and NDAVI, showing increase sediments causes decrease in NDAVI signals, and confirmed spatial relationship, where regions experiencing higher clustered turbidity samples showing more fragmented visible seagrass with the NDAVI index overlay. This was accurate to the expectation, although more suspended sediments is a stressor for seagrass, the results could also indicate a missed signal from sediment or debris blocking S2 from detecting seagrass with NDAVI, not necessarily no seagrass present.

Figure 10. Seasonal Index Comparison: Year-over-Year (2016-2024):

Overall trends show consistent convergence in summer, clustered around zero (-0.5-0.5), followed by divergence in fall/winter of NDVI, NDAVI, and SSSII. Alignment of indices represent peak growth in summer for vegetation, and natural seasonal variation of typical fall/winter environmental stressors like storms and more sediment (during Hurricane and peak season). The year 2017 shows a major divergence in indices followed by a “snap-back” to typical trends in 2018, indicating a resilient ecosystem in disaster recovery post Hurricane Irma.

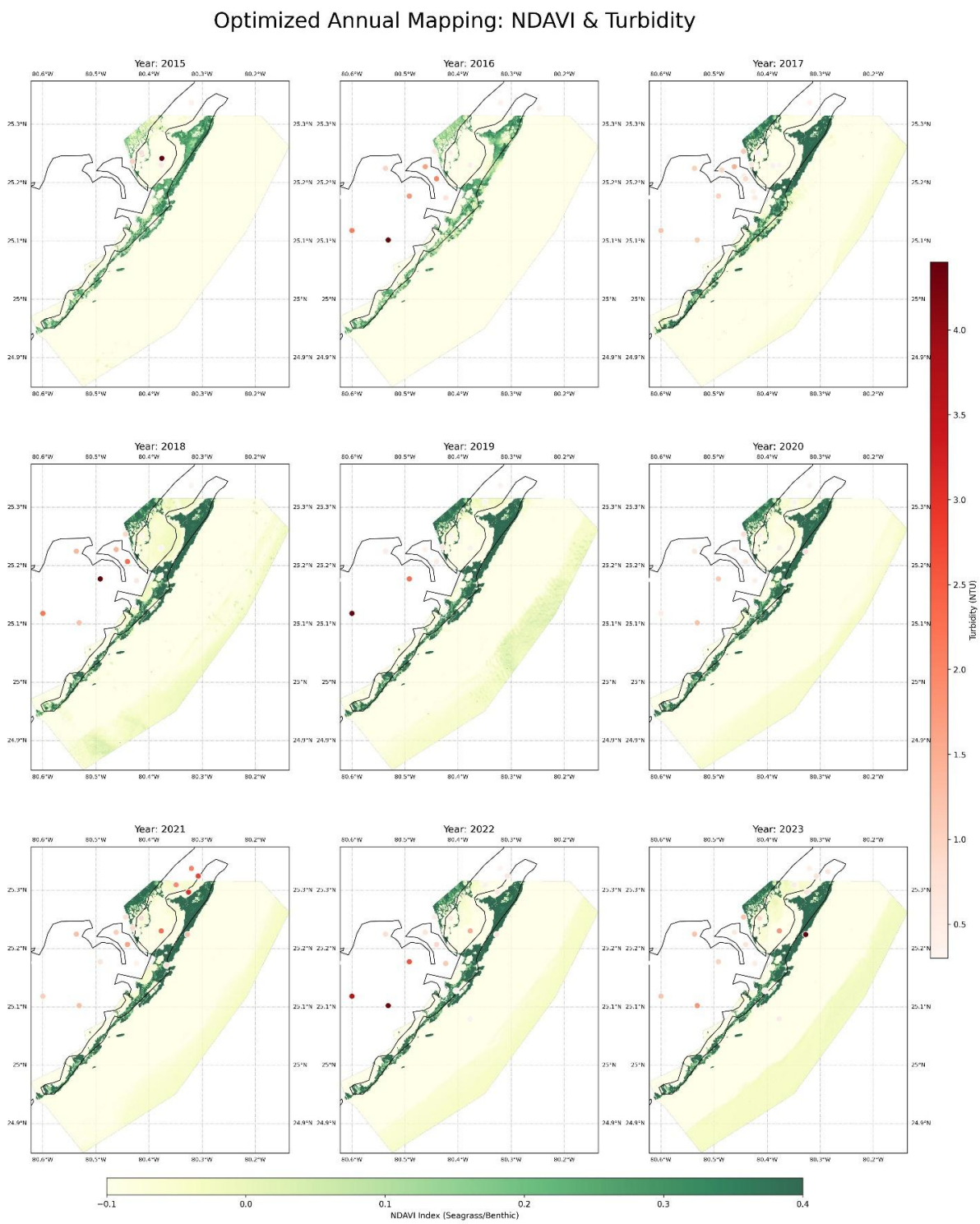


Figure 11. Annual Full Coverage Mapping: NDAVI vs. Turbidity:

High turbidity (red marker > 3.0 NTU) is spatially clustered along the northern coast of the keys. Shades of green (NDAVI) fluctuate the most along the northern coast as well. Therefore, validating turbidity levels cause stress to seagrass as hotspots of turbidity directly correspond to variable or fragmented seagrass distributions. Note: 2017 appears most fragmented and could be attributed to Hurricane Irma.

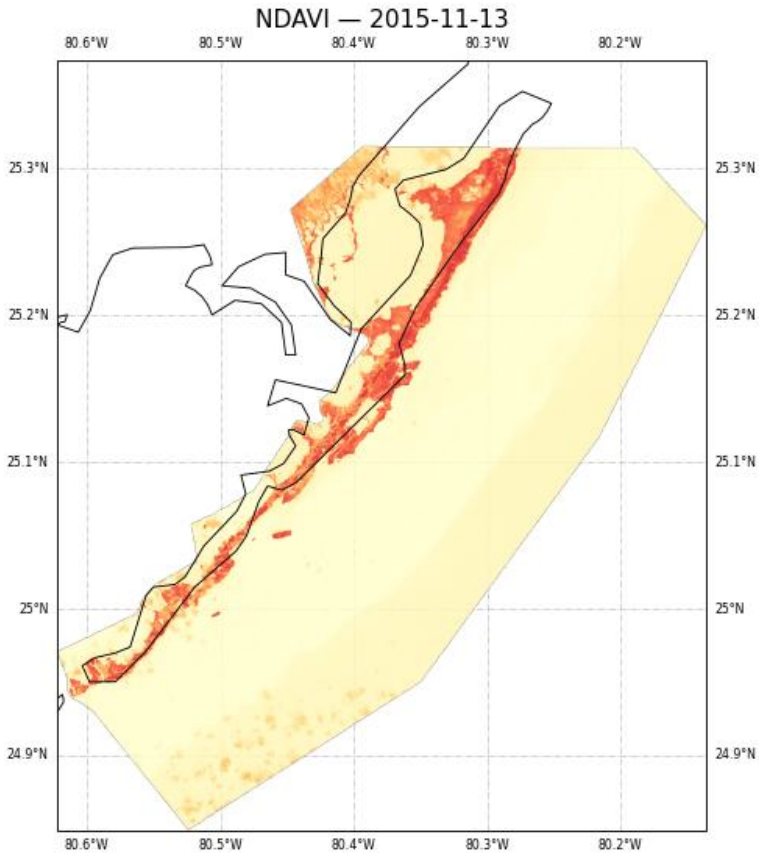


Figure 12. (Animation) NDAVI – 2015-2024:

The animated map projection shows a "black-out" of NDAVI signal at the end of 2017. Combined with results in the seasonal index comparison (fig 10), the "black-out" may further indicate the ecosystem under stress due to Hurricane Irma followed by a resilient "snap-back" to healthy trends. There is most likely a combination of lost satellite signal and loss of seagrass to explain the "black-out" and major divergence. Overall, there seems to be growth over the study period with NDAVI becoming denser over time with natural variation as to be expected as well.

4. Conclusion

Overall, our analysis confirms that seagrass distribution is strongly influenced by water clarity and environmental conditions. Spatial patterns in aquatic vegetation indices (NDAVI) show consistently higher values in regions of lower turbidity, reinforcing light availability as a primary factor for seagrass health in coastal systems. Time-series (Figure 6) analysis revealed significant trends in both Secchi Depth ($\tau = 0.163$, $p = 0.009$) and NDAVI ($\tau = 0.162$, $p = 0.031$) over the study period, indicating gradual improvement in water clarity and spectral seagrass detection. Anomaly analysis (Figure 8) shows a transition from predominantly negative to positive

deviations in both parameters following Hurricane Irma (2017), marking a potential ecological turning point possibly linked to sediment redistribution. Integrating satellite-derived indices with in-situ water quality data provides a capable framework for assessing coastal ecosystem dynamics over time.

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